

Trainings ▾

Test

Turn the keyswitch to the ON position.

Connect the ICON™ electronic service tool (Aftermarket or OEM systems), or INSITE™ electronic service tool (Integrated systems).

Initiate the Alarm Test.

If the alarm sounds, the alarm circuit passed the test and no repairs are needed.

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Resistance Check

Turn the keyswitch to the OFF position.

Remove and inspect the power fuse on the ICON™ engine harness for corrosion, damage, or a blown fuse.

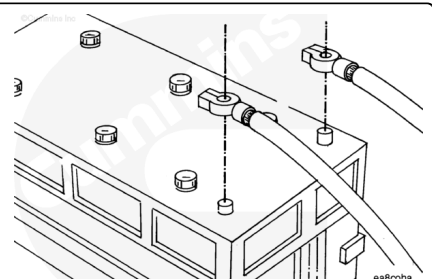
Replace the fuse if necessary.



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⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (--) battery cable last.

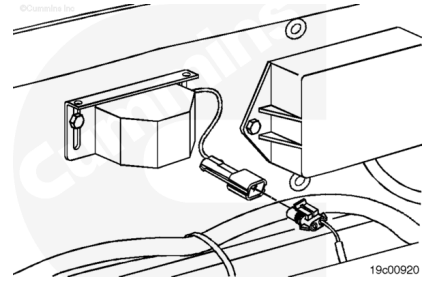


Disconnect the vehicle battery cables from the battery.
Remove the ICON™ engine harness positive (+) battery connector from the battery terminal post.

Note : Disconnecting the vehicle battery connections can require that the engine ECM real-time clock be reset using INSITE™.

⚠ CAUTION ⚠

To avoid pin and harness damage, use test leads, Part Number 3822758 and 3822917, when taking a measurement.



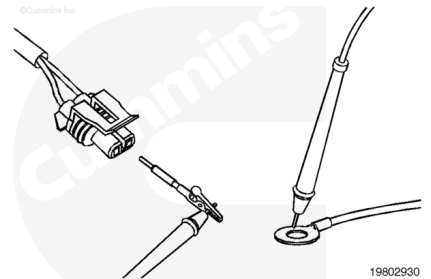
Disconnect the engine start alarm connector from the ICON™ engine harness. Disconnect the ICON™ idle control module A and B connectors.

Set the multimeter to measure resistance.

Touch one of the multimeter leads to the positive (+) ICON™ engine harness connector ring terminal.

Touch the other multimeter lead to pin B of the engine start alarm harness connector.

Read the value displayed on the multimeter.

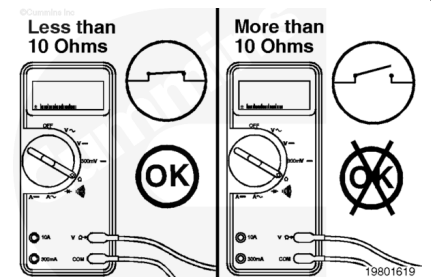


The multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

If the circuit is **not** closed, repair or replace the ICON™ engine harness.

Refer to Procedure 019-197

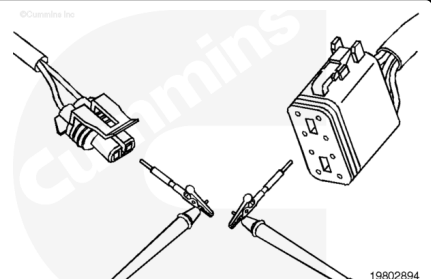
(/qs3/pubsys2/xml/en/procedures/99/99-019-197.html), 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html), or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin 1 of the ICON™ idle control module A harness connector.

Touch the other multimeter lead to pin B of the engine start alarm harness connector.

Read the value displayed on the multimeter.

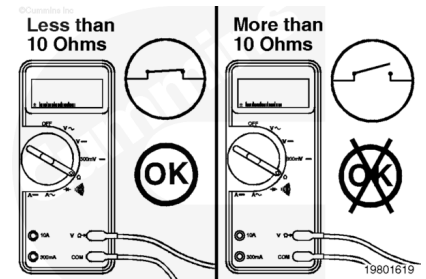


The multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

If the circuit is **not** closed, repair or replace the ICON™ engine harness.

Refer to Procedure 019-206

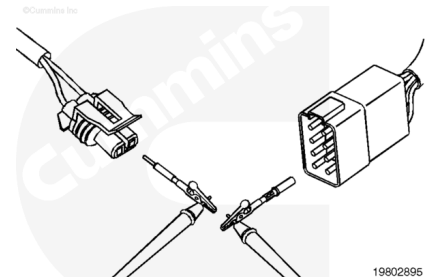
(/qs3/pubsys2/xml/en/procedures/99/99-019-206.html), 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html), or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin 3 of the ICON™ idle control module B harness connector.

Touch the other multimeter lead to pin A of the engine start alarm harness connector.

Read the value displayed on the multimeter.



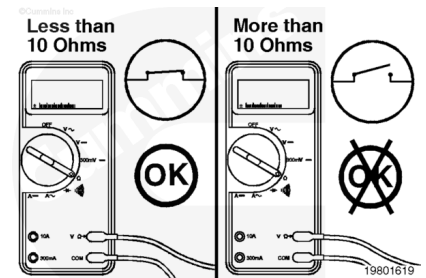
The multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

If the circuit is **not** closed, repair or replace the ICON™ engine harness.

Refer to Procedure 019-206

(/qs3/pubsys2/xml/en/procedures/99/99-019-206.html), 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html), or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

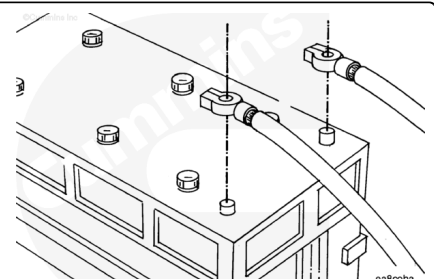
Connect all components after completing the repair.



Check for Short Circuit to Ground

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Turn the keyswitch to the OFF position. Disconnect the vehicle battery cables from the battery. Remove the ICON™ engine harness positive (+) battery connector from the positive (+) battery terminal post.

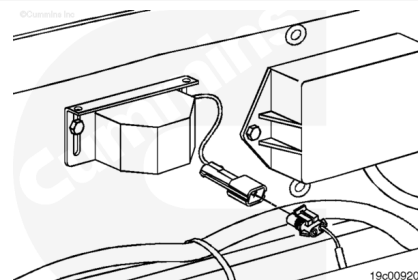
Note : Disconnecting the vehicle battery connections can require that the engine ECM real-time clock be reset using INSITE™.

⚠ CAUTION ⚠

To avoid pin and harness damage, use test leads, Part Number 3822758 and 3822917, when taking a measurement.

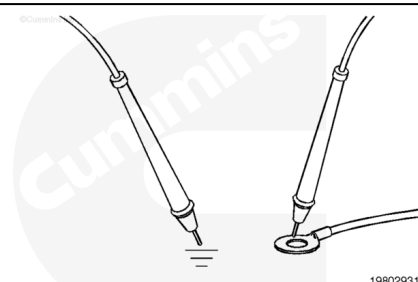
Disconnect the engine alarm from the ICON™ engine harness.
Disconnect the ICON™ idle control module A and B connectors.

Set the multimeter to measure resistance.



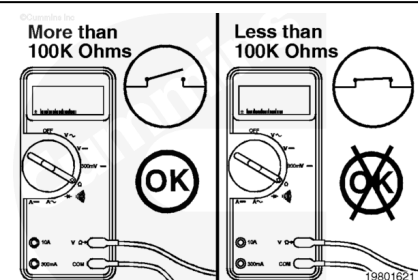
Touch one of the multimeter leads to the ICON™ engine harness positive (+) battery ring terminal connector. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.



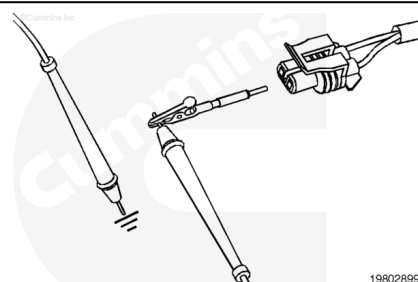
The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit to ground.
Repair or replace the ICON™ engine harness. Refer to Procedure 019-197 (/qs3/pubsys2/xml/en/procedures/99/99-019-197.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin A of the engine alarm harness connector. Touch the other multimeter lead to ground. Read the value displayed on the multimeter.

Then, touch one of the multimeter leads to pin B of the engine alarm harness connector. Touch the other multimeter lead to ground. Read the value displayed on the multimeter.

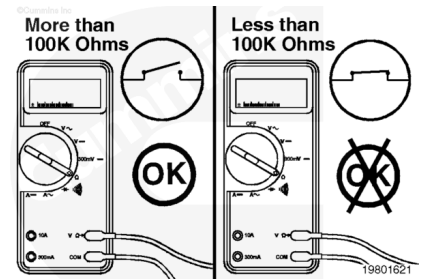


For both pin-to-ground checks, the multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit to ground.
Repair or replace the ICON™ engine harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-

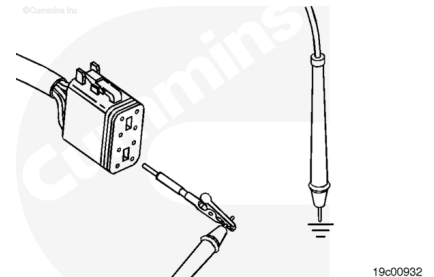
019-202.html) or 019-043

(/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



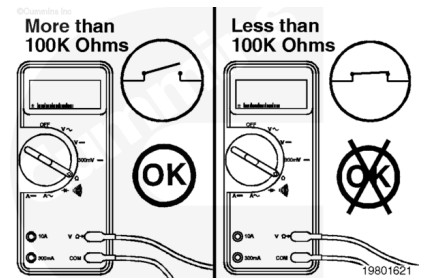
Touch one of the multimeter leads to pin 1 of the ICON™ idle control module A harness connector. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.



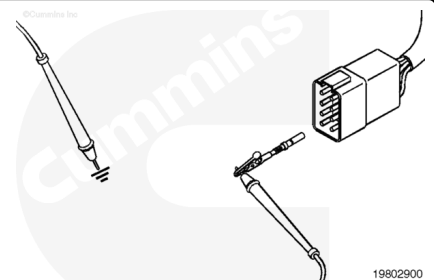
The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit to ground. Repair or replace the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin 3 of the ICON™ idle control module B harness connector. Touch the other multimeter lead to ground.

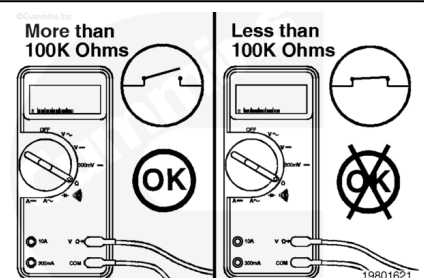
Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit to ground. Repair or replace the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

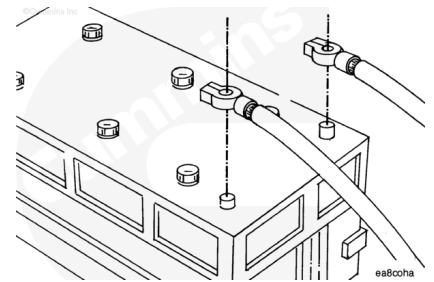
Connect all components after completing the repair.



Check for Short Circuit from Pin to Pin

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



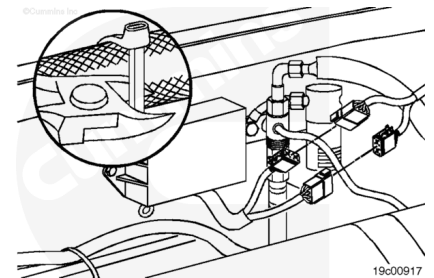
Turn the keyswitch to the OFF position. Disconnect the vehicle battery cables from the battery. Remove the ICON™ engine harness positive (+) battery connector from the positive (+) battery terminal post.

Set the multimeter to measure resistance.

Note : Disconnecting the vehicle battery connections can require that the engine ECM real-time clock be reset using INSITE™.

⚠ CAUTION ⚠

To avoid pin and harness damage, use test leads, Part Number 3822758 and 3822917, when taking a measurement.



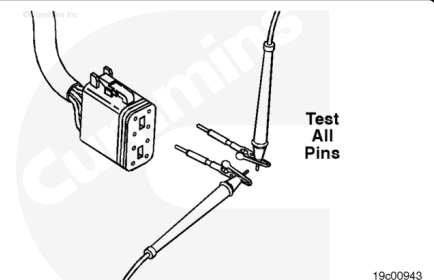
Disconnect the ICON™ idle control module A and B connectors.

Disconnect the engine start alarm from the ICON™ engine harness.

Touch one of the multimeter leads to pin 1 of the ICON™ idle control module A harness connector.

Touch the other multimeter lead to all other pins in the connector.

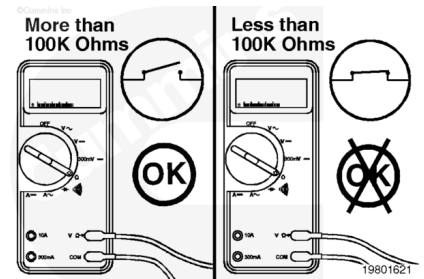
Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit from pin 1 in the ICON™ idle control module A harness connector to any other pin in the connector that registered a closed circuit.

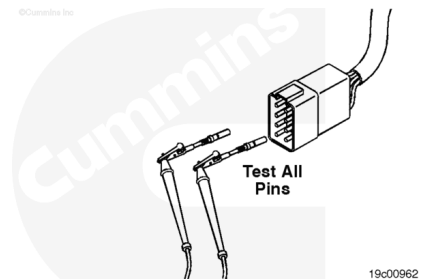
Repair or replace the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin 3 of the ICON™ idle control module B harness connector.

Touch the other multimeter lead to all other pins in the connector.

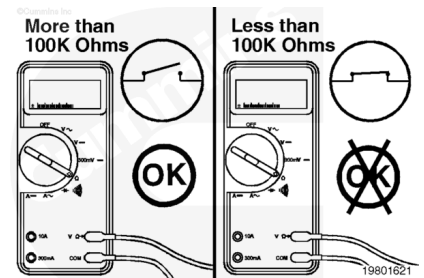
Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit from pin 3 in the ICON™ idle control module B harness connector to any other pin in the connector that registered a closed circuit.

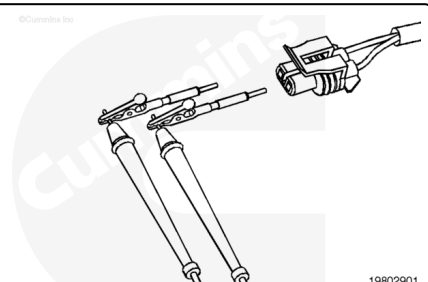
Repair or replace the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Touch one of the multimeter leads to pin A of the engine start alarm harness connector.

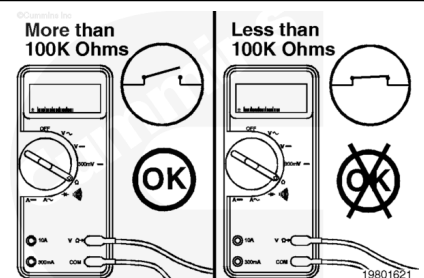
Touch the other multimeter lead to pin B of the connector.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit from pin A to pin B in the engine start alarm harness connector. Repair or replace the ICON™ engine harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).



Connect all components after completing the repair.

Check for Short Circuit to External Voltage Source

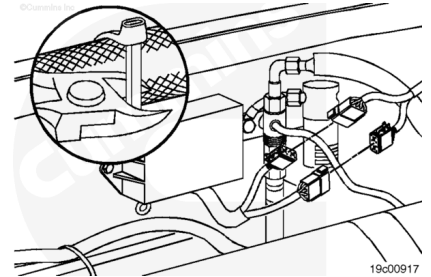
⚠ CAUTION ⚠

To avoid pin and harness damage, use test leads, Part Number 3822758 and 3822917, when taking a measurement.

Disconnect the ICON™ engine harness ring terminal positive (+) battery connector from the positive (+) battery post.
Disconnect the ICON™ idle control module A and B connectors.
Disconnect the engine start alarm from the ICON™ engine harness.

Set the multimeter to measure VDC.

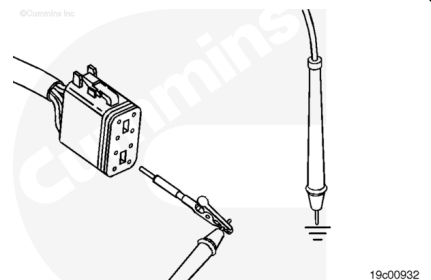
Turn the keyswitch to the ON position.



Touch one of the multimeter leads to pin 1 of the ICON™ idle control module A harness connector.

Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

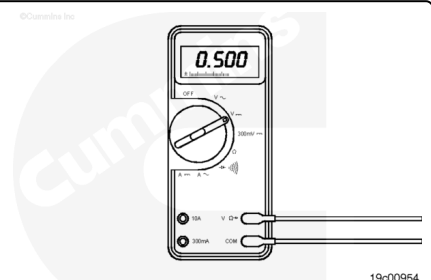


The multimeter **must** display a reading of less than 0.5 VDC.

If the voltage is **not** less than 0.5 VDC, there is a short circuit to an external voltage source.

Note : An external voltage source is any wire in the cab harness or engine harness wiring that carries voltage.

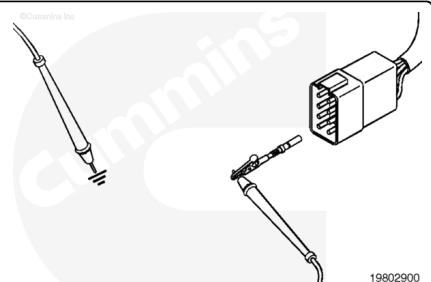
Remove the external voltage source.



Touch one of the multimeter leads to pin 3 of the ICON™ idle control module B harness connector.

Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

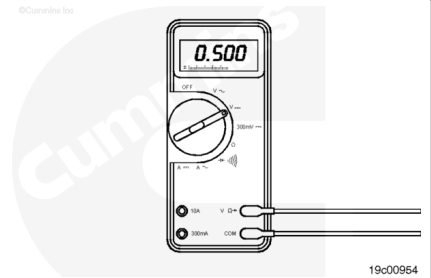


The multimeter **must** display a reading of less than 0.5 VDC.

If the voltage is **not** less than 0.5 VDC, there is a short circuit to an external voltage source.

Note : An external voltage source is any wire in the cab harness or engine harness wiring that carries voltage.

Remove the external voltage source.

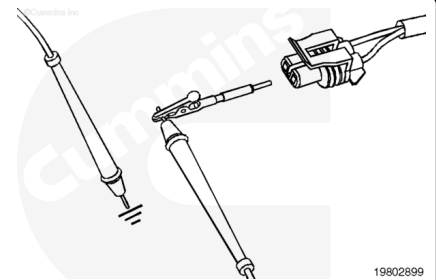


Touch one of the multimeter leads to pin A of the engine start alarm harness connector. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.

Repeat the pin-to-ground check from pin B. Touch one of the multimeter leads to pin B of the engine start alarm harness connector. Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.



For both checks, the multimeter **must** display a reading of less than 0.5 VDC.

If the voltage is **not** less than 0.5 VDC, there is a short circuit to an external voltage source.

Note : An external voltage source is any wire in the cab harness or engine harness wiring that carries voltage.

Remove the external voltage source.

Connect all components after completing the repair.

